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CASE STUDY

OEE Validation & Downtime Reduction

with MES in Automotive Manufacturing

Precision Machining & Injection Molding | 3 Plants | 120+ Machines

3

Plants

120+

Machines

Tier-2

Supplier

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Business Challenge

Manual OEE

Spreadsheet-based calculations with no real-time visibility of machine performance

Untracked Downtime

Incorrectly categorized downtime and frequent discrepancies in data

Reactive Maintenance

No proactive planning; breakdowns handled after-the-fact

No Visibility

Limited shop floor transparency and lack of accountability

Data Errors

Availability, performance, and quality data riddled with manual errors

Low Utilization

Machine utilization stuck below 60%, causing production delays

Solution Implemented

MES with Real-Time Machine Monitoring & OEE Validation

IoT Integration

Automatic tracking of machine status: Run, Idle, Breakdown

Automated OEE

Availability, Performance & Quality calculated in real time

Downtime Classification

Predefined and customizable downtime categories

Andon Alerts

Instant escalation alerts for machine stoppages

Live OEE Dashboard

Shift-wise, machine-wise, and plant-wise visibility

Root Cause Analysis

Drill-down reports for downtime and production losses

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Implementation Approach

Phase 01

Baseline Study

- Captured existing OEE & downtime data
- Identified major loss areas

Phase 02

Pilot Implementation

- Deployed MES on critical machines
- Validated real-time OEE accuracy

Phase 03

Full-Scale Rollout

- Connected all machines across plants
- Standardized downtime codes & processes

Phase 04

Training & Adoption

- Trained operators, supervisors & maintenance
- Introduced accountability dashboards

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Results & Impact

58%→78%

OEE Improvement

in 6 months

35%

Downtime Reduction

unplanned downtime cut

20%

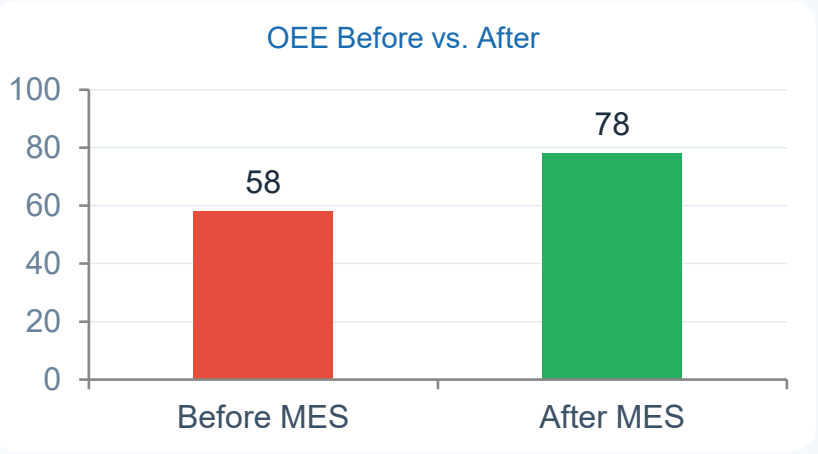
Higher Utilization

machine efficiency gain

100%

OEE Accuracy

real-time automated data



Key Benefits

- Elimination of manual data errors
- Standardized OEE across all plants
- Faster decisions with live dashboard data
- Improved preventive maintenance planning
- Data-driven continuous improvement

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Before vs. After Snapshot

Metric	Before MES	After MES
OEE	58%	**78%**
Downtime	High (Uncontrolled)	**Reduced by 35%**
Data Accuracy	Low (Manual)	**High (Automated)**
Visibility	Limited	**Real-Time**
Decision Making	Reactive	**Proactive**
Maintenance	Unplanned / Reactive	**Preventive / Planned**

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Conclusion

From Estimated to Exact: Precision Manufacturing Intelligence

By implementing an MES-driven OEE validation system, the organization transitioned from estimated performance tracking to precise, real-time monitoring. This enabled significant downtime reduction, improved machine efficiency, and stronger operational control — creating a measurable competitive advantage in automotive manufacturing.

OEE: +20pts	**Downtime: -35%**	**Utilization: +20%**	**Data: 100% Accurate**

Tier-2 Automotive Manufacturer | MES Implementation Case Study